

What VoIP means to businesses?

How VoIP works for a business is simple: It adds voice to a data network to reduce costs, improve productivity, and enhance collaboration. VoIP and IP telephony are becoming increasingly popular with large corporations and consumers alike.

For many people, Internet protocol (IP) is not only a way to transport data but also a tool that simplifies and streamlines a wide range of business applications, telephony being the most obvious example. VoIP is also the foundation for more advanced unified communications applications—including Web and video conferencing—that can transform the way organisations do business. The benefits include:

1. An organisation saves money by having one network to manage instead of two.
2. Organisations can easily add, move or change phone extensions and locations, which saves money and gives more flexibility.
3. Wireless IP phones connect users to the communications system and data resources, such as customer information, while they are in the warehouse, on the sales floor or anywhere they can access the organisation's data network wirelessly.
4. Unified communication solutions for small businesses go beyond basic VoIP capabilities to enhance collaboration. With a unified communications solution, workers can easily collaborate through voice, video chat, Web conference and instant messaging. Employees can collaborate using each technology individually or all of them simultaneously, and from a single, easy-to-use interface.
5. VoIP allows organisations to use features such as voicemail, caller ID, conferencing, forwarding and unlimited long-distance calls without having to pay extra for these.
6. Since VoIP is based on the Internet Protocol, it also handles other media types than voice such as transferring images, video and text. It is possible to share files or even communicate with the use of a Web cam.

—By Minhaj Zia, national sales manager-Unified Communications, Cisco, India & SAARC

their circuit-switched voice revenue streams. Moreover, cellular operators face intense competition from the more popular Web-based VoIP alternatives that are permeating the mass market.

"Mobile VoIP is still in the infancy as the mobile networks that can support end-to-end IP telephony are still in very early stages of build-out. Moreover, the traditional cellular networks still have a long life left in them, which will prevent large rollouts of IP-based mobile networks," comments Sharma.

But once it picks up, mVoIP has the potential to play a crucial role in enhancing the appeal of VoIP not just for large and small enterprises but also for individuals, who could use this technology to stay in touch with their family and friends residing in India and abroad. After all, the Indian population does have a penchant for all things mobile!

VoIP in India

VoIP remains subject to several regulations by the Department of Telecom-

munications (DoT) and the Telecom Regulatory Authority of India (TRAI). There was a time when it was seen as completely illegal (it continues to be so in several other countries), but now authorised service providers are allowed to provide VoIP services to the public. The legality of VoIP remains a big question in several countries because there is no regulatory framework, no uniformity in services and standards, and no way to clearly differentiate information services from telecommunications! All of which makes life very difficult for the law-makers.

A few years ago, VoIP got its big break in India when the TRAI gave its approval for the logical partitioning of PSTN and closed user group (CUG) networks. Standards such as the session initiation protocol (SIP) also emerged to enable integration of legacy or hybrid systems with newer IP-based ones. As a result, companies could merge their voice and data networks and reduce their investments therein.

However, Minhaj Zia, national

sales manager - Unified Communications, Cisco, India & SAARC, points out that according to TRAI guidelines, VoIP can only be provided within CUGs in the country. Hence such services provided over the PSTN would be classified as illegal. As a result, although VoIP has picked up at the enterprise level, it is yet to make its way into the consumer household, and might not do so in the near future.

"In a wire-line home environment with standard copper connections like in India, VoIP can be provided only on PCs associated with digital subscriber lines (DSL). I believe that number is only a few millions. (The total copper connections in India are approximately 35 million and most of them do not meet the minimum quality requirements for broadband connections.) Many times, the quality of copper connections do not meet appropriate standards and hence DSL cannot even be provided. Hence VoIP becomes a non-starter," adds Dr Borkar.

Srinivasan of VIVA Communications, which offers DoT-authorized VoIP services, explains that VoIP has largely been evaded by small- and medium-sized businesses (SMBs) and has remained restricted to large enterprises, mainly due to the cost of IP phones and lack of effective penetration of broadband services in tier II and III cities. Plus, even though VoIP is considerably cheaper than the traditional telephone service, it has remained attractive only to those segments of the market which spend a lot on international calls.

The above trends ought to reverse before VoIP becomes more popular in India.

SIP trunking: 1 + 1 = 3

The adoption of VoIP in developed countries is increasing now, partly due to technologies like SIP—a trend that might extend to India. In those countries too, the widespread existence of legacy circuit-switched networks was holding back operators from moving to VoIP. But, with advent of SIP, the